

Q & A Part 2

Brainster Web Development Academy



Exercise 1

Create a "To-Do List" application.

1. Design an HTML page with an input field and a button to add tasks to a list of results.
2. Write JavaScript code that adds the task to the list when the button is clicked.



Solution 1

3

```
const button = document.querySelector("#btnAdd");
const list = document.querySelector("#list");
const input = document.querySelector("input");

button.addEventListener("click", () => {
  console.log("clicked button");
  const value = input.value;
  list.innerHTML += `<li>${value}</li>`;
  input.value = "";
});
```



Solution 2

3

```
const list = document.querySelector("#list");
const form = document.querySelector("form");

form.addEventListener("submit", (event) => {
  event.preventDefault();
  const value = event.target["todo"].value;
  list.innerHTML += `<li>${value}</li>`;
  event.target.reset();
});
```



Exercise 2

Create a basic calculator.

1. Design an HTML page with input fields for two numbers and buttons for addition, subtraction, multiplication, and division.
2. Write JavaScript code that performs the corresponding operation when the user clicks the respective button and displays the result on the page.



Solution

```
const additionBtn = document.querySelector("#addition");
const subtractionBtn = document.querySelector("#subtraction");
const multiplicationBtn = document.querySelector("#multiplication");
const divisionBtn = document.querySelector("#division");
```

```
const addition = (num1, num2) => num1 + num2;
const subtraction = (num1, num2) => num1 - num2;
const multiplication = (num1, num2) => num1 * num2;
const division = (num1, num2) => num1 / num2;
```

```
const input1 = document.querySelector("#number1");
const input2 = document.querySelector("#number2");
const resultParagraph = document.querySelector("#result");
```

```
additionBtn.addEventListener("click", () => {
  const number1 = +input1.value;
  const number2 = +input2.value;
  let result = addition(number1, number2);
  resultParagraph.textContent = `${result}`;
});
```

```
subtractionBtn.addEventListener("click", () => {
  const number1 = +input1.value;
  const number2 = +input2.value;
  let result = subtraction(number1, number2);
  resultParagraph.textContent = `${result}`;
});
multiplicationBtn.addEventListener("click", () => {
  const number1 = +input1.value;
  const number2 = +input2.value;
  let result = multiplication(number1, number2);
  resultParagraph.textContent = `${result}`;
});
divisionBtn.addEventListener("click", () => {
  const number1 = +input1.value;
  const number2 = +input2.value;
  let result = division(number1, number2);
  resultParagraph.textContent = `${result}`;
});
```



Exercise 3

Create a "Rock, Paper, Scissors" game.

1. Design an HTML page with buttons for the player to choose rock, paper, or scissors.
2. Write JavaScript code that generates a random choice for the computer, compares it with the player's choice, and displays the result (win, lose, or draw) on the page.



Solution

```
const rockString = "rock";
const paperString = "paper";
const scissorsString = "scissors";

const gameValues = [rockString, paperString, scissorsString];

const rock = document.querySelector("#rock");
const paper = document.querySelector("#paper");
const scissors = document.querySelector("#scissors");

function game(userChoice) {
  const randomIndex = Math.floor(Math.random() * 3);
  const randomValue = gameValues[randomIndex];
  if (randomValue === userChoice) {
    console.log("draw", randomValue, userChoice);
  } else if (
    (randomValue === rockString && userChoice === scissorsString) ||
    (randomValue === "paper" && userChoice === "rock") ||
    (randomValue === "scissors" && userChoice === "paper")
  ) {
    console.log("you lose!", randomValue, userChoice);
  } else {
    console.log("you win!", randomValue, userChoice);
  }
}

rock.addEventListener("click", () => {
  game(rockString);
});
paper.addEventListener("click", () => {
  game(paperString);
});
scissors.addEventListener("click", () => {
  game(scissorsString);
});
```



Exercise 4

Form Validation

1. Create an HTML page with a form that allows the user to enter their email address and password.
2. When the form is submitted, use JavaScript to validate the inputs.
3. Display an error message if the name field is empty or the email address is not in a valid format.
4. If both inputs are valid, display a success message.



Solution

3

```
const emailFormat = /^\\w+([\\.-]?\\w+)*@\\w+([\\.-]?\\w+)*\\.\\w{2,3})+$/;  
const emailFormat2 =  
  
/^(((\\[<>()\\[\\]\\\\.,;:\\s@"]+(\\[<>()\\[\\]\\\\.,;:\\s@"]+)*)(\\.[^."<>()\\[\\]\\\\.,;:\\s@"]+))@(((\\[[0-9]{1,3}\\.[0-9]{1,3}\\.[0-9]{1,3}\\.[0-9]{1,3})|(((a-zA-Z\\-0-9)+\\.)+[a-zA-Z]{2,})))$)/;  
  
const form = document.querySelector("form");  
const errorParagraph = document.querySelector("#error");  
const successParagraph = document.querySelector("#success");  
  
form.addEventListener("submit", validateForm);  
  
...  
  
...  
  
...
```



Solution

3

```
function validateForm(event) {  
  event.preventDefault();  
  const email = event.target["email"].value.trim();  
  const password = event.target["password"].value.trim();  
  
  if (email === "" || password === "") {  
    successParagraph.style.display = "none";  
    errorParagraph.style.display = "block";  
    errorParagraph.textContent = "Please fill all the fields!";  
    errorParagraph.style.color = "red";  
  } else if (!email.match(emailFormat)) {  
    successParagraph.style.display = "none";  
    errorParagraph.style.display = "block";  
    errorParagraph.textContent = "Please enter a valid email address!";  
  } else {  
    errorParagraph.style.display = "none";  
    successParagraph.textContent = "You are logged in!";  
    successParagraph.style.color = "green";  
    event.target.reset();  
  }  
}
```

